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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

8449417321

PHYSICS **5054/33**

Paper 3 Practical Test **May/June 2010**

2 hours

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
For each of the questions in Section A, you will be allowed to work with the apparatus for a maximum of 20 minutes. For the question in Section B, you will be allowed to work with the apparatus for a maximum of 1 hour.
You are expected to record all your observations as soon as these observations are made.
An account of the method of carrying out the experiments is **not** required.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
Total	

This document consists of **10** printed pages and **2** blank pages.

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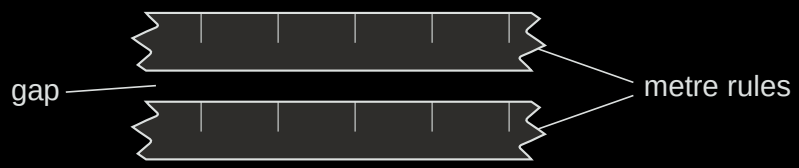


Fig. 1.1 (top view)

Ensure that the spheres are touching each other and are in a line.

Measure the total length *L* occupied by the 20 spheres.

L = ... [1]

(b) State, with the aid of a diagram, a technique that you used to ensure that *L* was determined as accurately as possible.

.....

..... [1]

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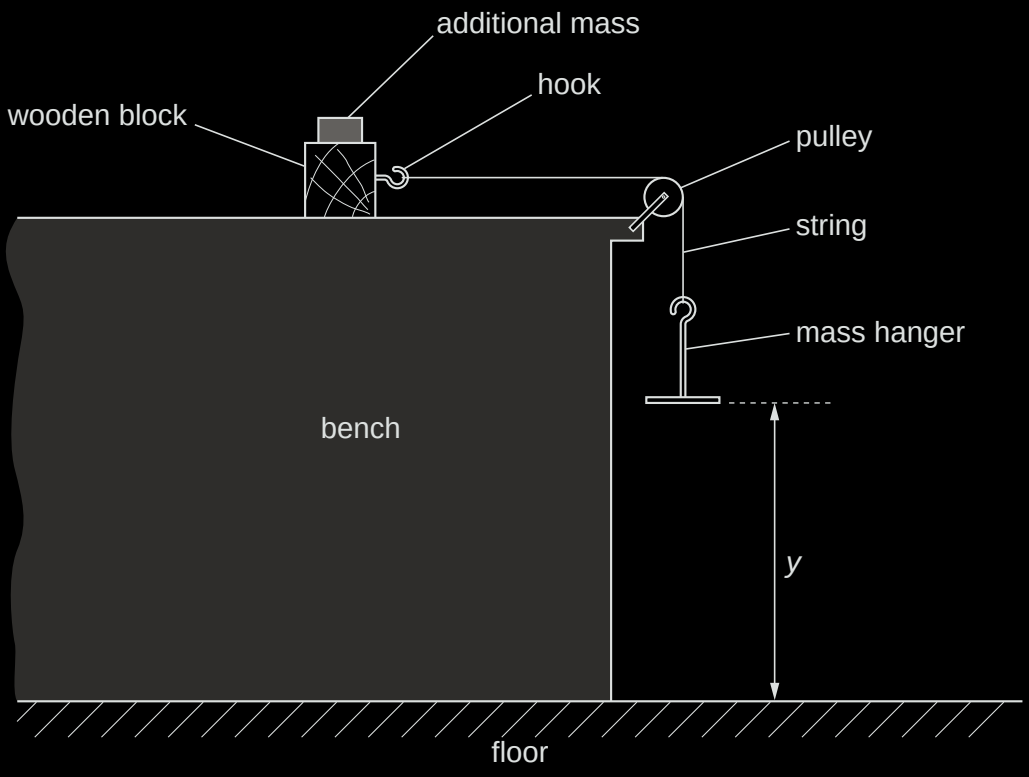


Fig. 2.1

Move the block away from the pulley until the mass hanger is just below the pulley. Clamp the metre rule vertically close to, but not touching, the mass hanger.

Measure the distance *y* from the base of the mass hanger to the floor.

y = ... [1]

(b) Explain, with the aid of a diagram, how you ensured that the metre rule was vertical.

.....

..... [1]

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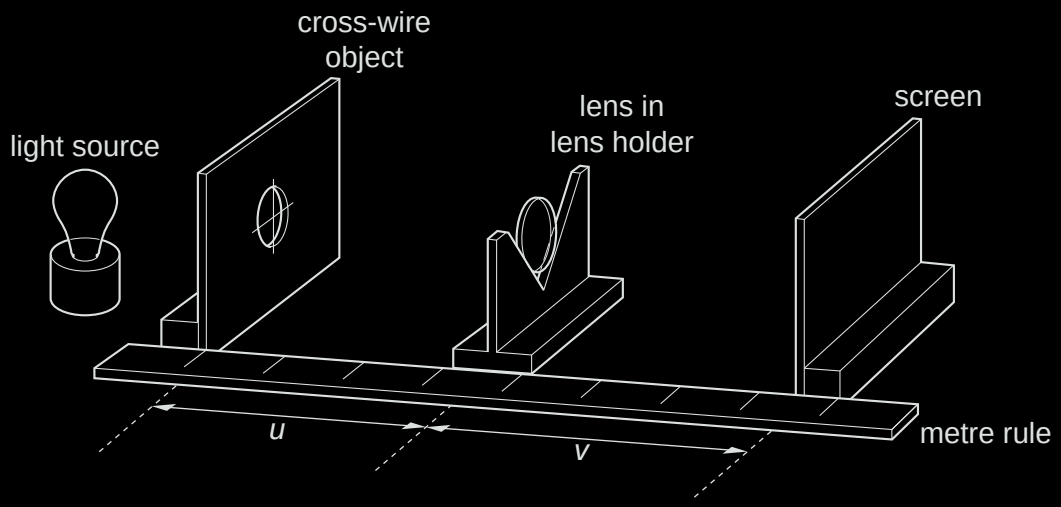


Fig. 3.1

Adjust the position of the lens until the distance *u* between the object and the lens is 20.0 cm. Adjust the position of the screen until a sharply-focused image of the cross-wire object is formed on the screen.

Determine

(i) the distance *v* between the lens and the screen,

v = ...

(ii) the average diameter *D* of the image formed on the screen.

D = ...

[2]

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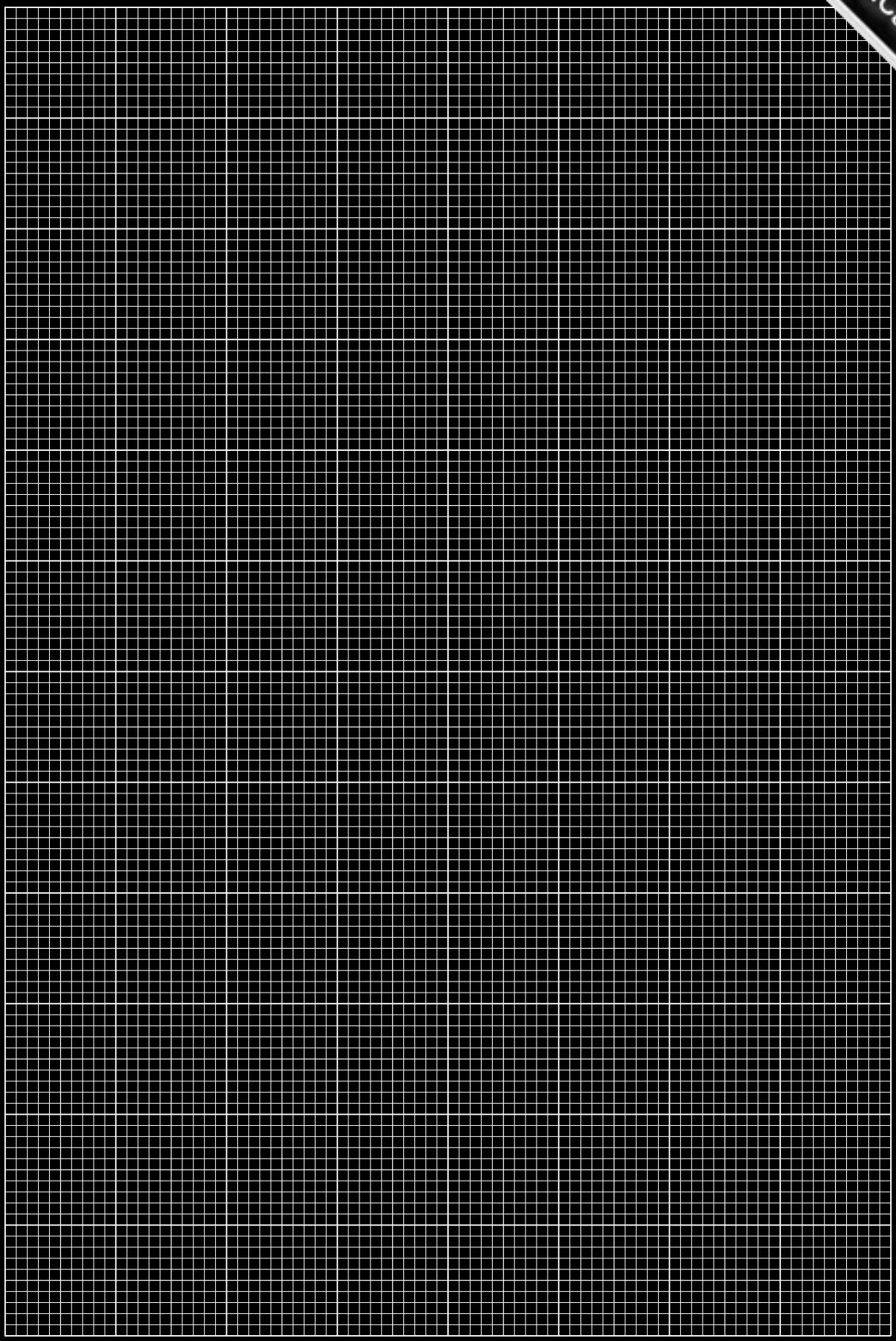
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